Homework #7

T3, 4, 12, 19, 25, 26, 34, 35

T3 $a \le b \Leftrightarrow a \vee b = b$, $b \le c \Leftrightarrow b = b \wedge c$

故 $a \vee b = b = b \wedge c$

左式 $= (a \wedge b) \vee (b \wedge c) = a \vee b = b$

右式 $= (a \vee b) \wedge (b \vee c) = b \wedge c = b$

故 $(a \wedge b) \vee (b \wedge c) = (a \vee b) \wedge (b \vee c)$

T4 证明: $(a \wedge b) \vee (c \wedge d)$

$\le (a \wedge b \vee c) \wedge (a \wedge b \vee d)$

$\le (a \vee c) \wedge (b \vee c) \wedge (a \vee d) \wedge (b \vee d)$

$\le (a \vee c) \wedge (b \vee d)$

T12 证明: 由等幂律 $L \vee L = L$, $L \wedge L = L$

若 L 是自身的补, $L \vee \bar{L} = 1$, $L \wedge \bar{L} = 0 \Rightarrow L \vee L = 1$, $L \wedge L = 0$

故 $L = 1 = 0$, 矛盾

故不会有元素是自身的补.

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T19 假设当 $a=b_i$ $(1\le i\le r)$ 时，$a\le b_1\vee b_2\vee\cdots\vee b_r$

若 $a\le b_1\vee b_2\vee\cdots\vee b_r$，则 $a\wedge(b_1\vee b_2\vee\cdots\vee b_r)=a$ ——①

假设 $a\neq b_i$ $(1\le i\le r)$ 则 $a\wedge b_i=0$，$a\wedge(b_1\vee b_2\vee\cdots\vee b_r)=0$

$\Rightarrow a=0$，这与 a 是原子矛盾，故 $\exists i$，$a=b_i$

T5	x	y	$\bar{x}$	$\bar{y}$	$x\vee y$	$x\wedge(x\vee y)$	$\bar{x}\wedge\bar{y}$	$(x\wedge(x\vee y))\vee(\bar{x}\wedge\bar{y})$
	0	0	1	1	0	0	1	1
	0	α	1	β	α	0	β	β
	0	β	1	α	1	0	α	α
	0	1	1	0	1	0	0	0
	α	0	β	1	α	α	β	1
	α	α	β	β	α	α	β	1
	α	β	β	α	1	α	0	α
	α	1	β	0	1	α	0	α
	β	0	α	1	α	0	α	α
	β	α	α	β	α	0	0	0
	β	β	α	α	1	β	α	1
	β	1	α	0	1	β	0	β
	1	0	0	1	α	α	0	α
	1	α	0	β	α	α	0	α
	1	β	0	α	1	1	0	1
	1	1	0	0	1	1	0	1

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Date.

T26 $f(x,y) = \bigvee_{k=00}^{11}(c_k \wedge m_k)$

$= (f(1,1)\wedge x\wedge y)\vee(f(1,0)\wedge x\wedge \bar{y})\vee(f(0,1)\wedge \bar{x}\wedge y)\vee(f(0,0)\wedge \bar{x}\wedge \bar{y})$

$= (1\wedge x\wedge y)\vee(a\wedge x\wedge \bar{y})\vee(a\wedge \bar{x}\wedge y)\vee(1\wedge \bar{x}\wedge \bar{y})$

$f(x,y) = \bigwedge_{k=00}^{11}(c_k \vee m_k)$

$= (f(0,0)\vee x\vee y)\wedge[f(1,0)\vee\bar{x}\vee y]\wedge[f(0,1)\vee x\vee\bar{y}]\wedge[f(1,1)\vee\bar{x}\vee\bar{y}]$

$= (1\vee x\vee y)\wedge(a\vee\bar{x}\vee y)\wedge(a\vee x\vee\bar{y})\wedge(1\vee\bar{x}\vee\bar{y})$

T34 证明：$(a\vee b)\wedge(c\vee\bar{b})$

$= [a\wedge(c\vee\bar{b})]\vee[b\wedge(c\vee\bar{b})]$

$= (a\wedge c)\vee(a\wedge\bar{b})\vee(b\wedge c)\vee(\quad)$

$= (a\wedge\bar{b}\wedge c)\vee(a\wedge\bar{b})\vee(a\wedge b\wedge c)\vee(b\wedge c)$

$= (a\wedge\bar{b})\vee(c\wedge b)$

T35 证明：(1) $\forall x\in B_1$, $f(a\wedge x) = f(a)\otimes f(x) = f(a)$

故 $f(a) = a$, $a\in J$

(2) $\forall x\in B_1$, $f(a\wedge x) = a\otimes f(x) = a$

故 $\forall x\in B_1$, $a\wedge x\in J$.

取 $x=b$, 若 $b\le a$, 则 $a\wedge b = b\in J$

(3) $\forall x, y \in J,\ f(x \wedge y) = f(x) \otimes f(y) = \alpha \otimes \alpha = \alpha$

故 $x \wedge y \in J$

$f(x \vee y) = f(x) \oplus f(y) = \alpha \oplus \alpha = \alpha$

故 $x \vee y \in J$

故 $\langle J; \vee, \wedge \rangle$ 构成一代数系统

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